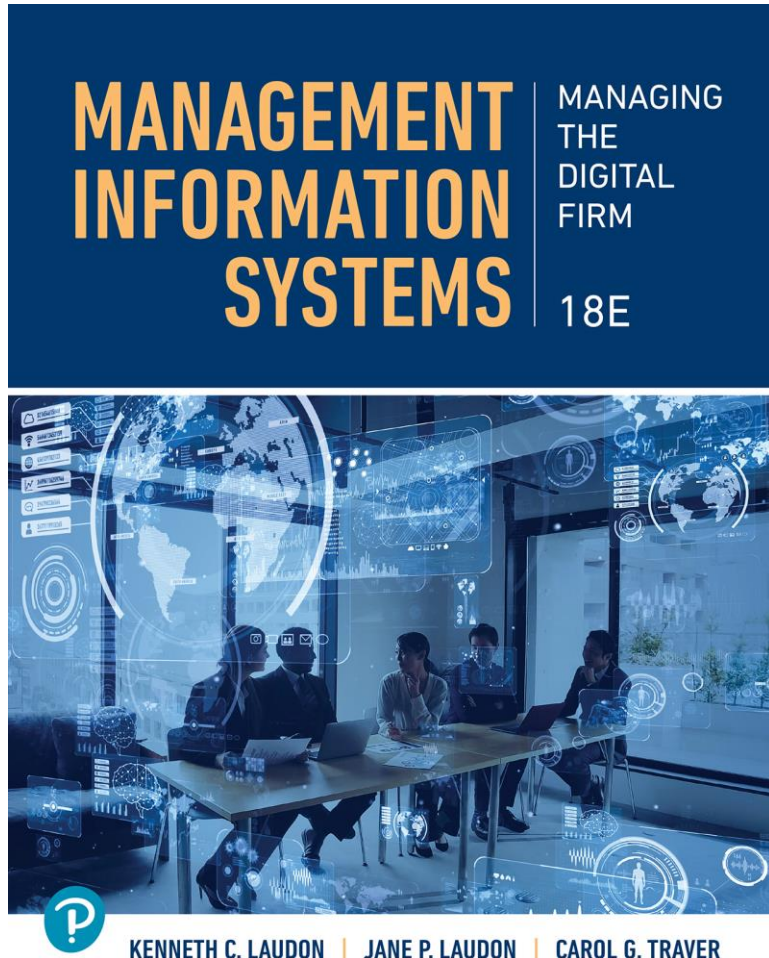


Management Information Systems: Managing the Digital Firm

Eighteenth Edition



Chapter 5

IT Infrastructure and Emerging Technologies

Learning Objectives (1 of 2)

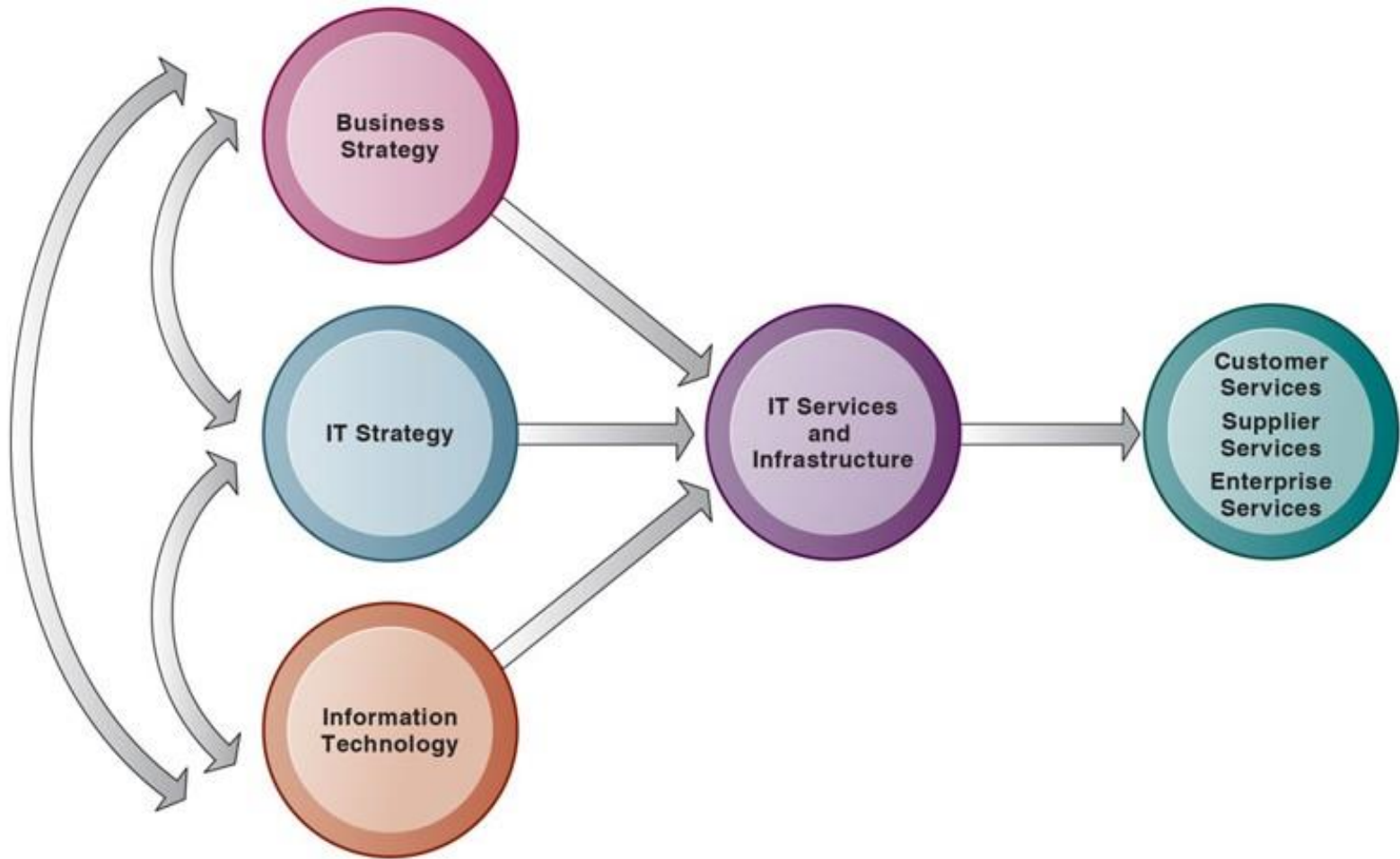
- 5.1** Discuss IT infrastructure and its evolution
- 5.2** Describe the components of IT infrastructure
- 5.3** Discuss the mobile platform, apps, and BYOD
- 5.4** Discuss quantum computing and green computing
- 5.5** Discuss virtualization and cloud computing
- 5.6** Describe contemporary software platforms
- 5.7** Discuss alternative ways of obtaining software
- 5.8** Describe IT infrastructure for AI
- 5.9** Discuss IT infrastructure management challenges
- 5.10** Understand how the information in this chapter can help your career

IT Infrastructure and Its Evolution

(1 of 6)

- A firm's IT infrastructure includes
 - Investment in hardware, software, and services
 - Consulting, education, and training, as well
 - Shared across an entire firm or across entire business units in the firm

Figure 5.1 Connections Between the Firm, IT Infrastructure, and Business Capabilities



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IT Infrastructure and Its Evolution

(2 of 6)

- Set of firmwide services composed of both human and technical capabilities include
 - Computing platforms
 - Telecommunications services
 - Data management services
 - Application software services
 - Physical facilities management services

IT Infrastructure and Its Evolution

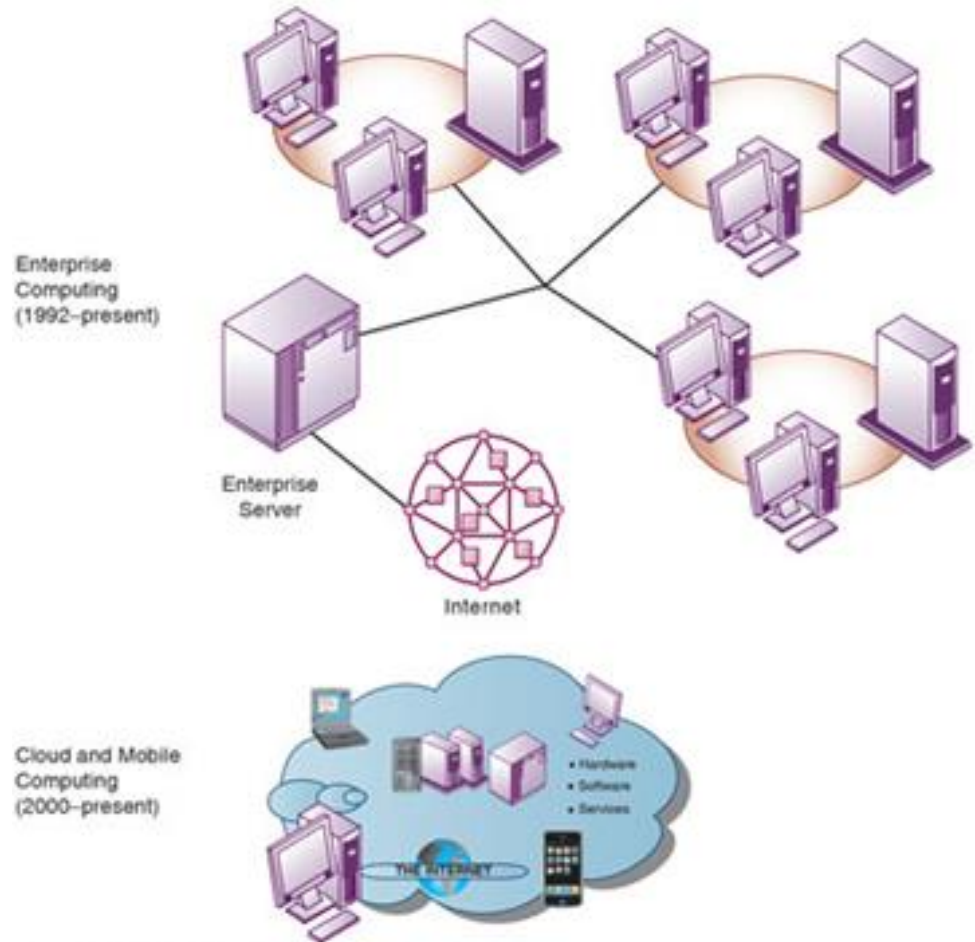
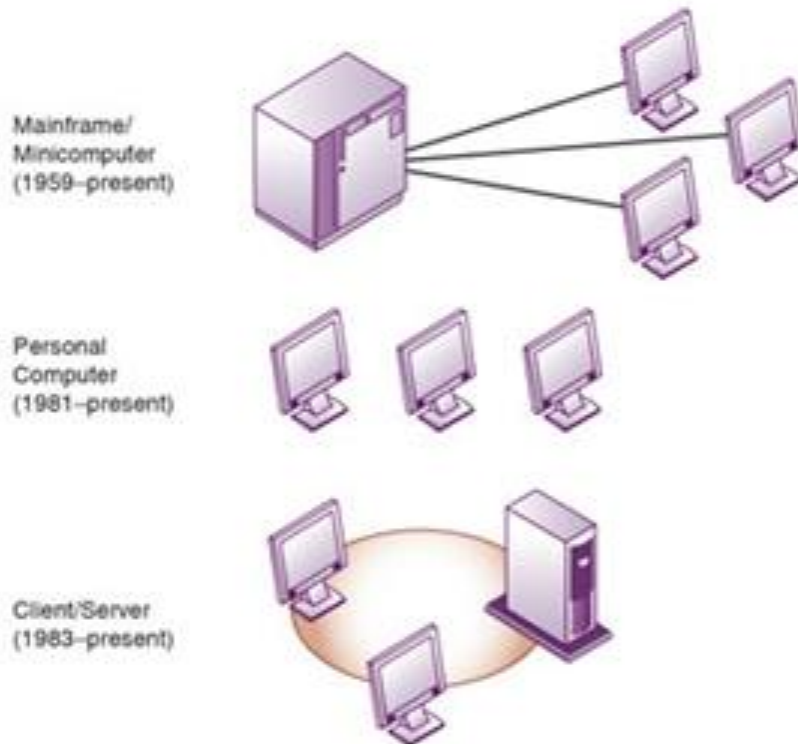
(3 of 6)

- Set of firmwide services composed of both human and technical capabilities include
 - IT management services
 - IT standards services
 - IT education services
 - IT research and development services

Figure 5.2

Eras in Infrastructure Evolution

Stages in IT Infrastructure Evolution



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IT Infrastructure and Its Evolution

(4 of 6)

Computing Era	Time Period	Infrastructure Description
Mainframe and Minicomputer Era	1959–present	Highly centralized computing under the control of professional programmers and systems operators, with most elements of infrastructure provided by a single vendor, the manufacturer of the hardware and the software, dominated by IBM. Smaller but powerful minicomputers support decentralized computing, customized to the specific needs of individual departments or business units. In recent years, the minicomputer has evolved into a midrange computer or midrange server and is part of a network.
Personal Computer Era	1981–present	Initially, small standalone desktop computers featuring desktop productivity software tools that were the first widely adopted by businesses and consumers. Primarily used Intel processors coupled with the Windows operating system.
Client/Server Era	1983–present	PCs called clients networked to powerful server computers with computer processing divided between both types of computers. Enables organizations to distribute computing work across the organization using a series of smaller inexpensive computers.
Enterprise Computing Era	1992–present	Network standards and protocols, including TCP/IP for the Internet, enable different pieces of computer hardware and smaller networks to link into an enterprise-wide network where information can flow freely across the organization and between the firm and other organizations.
Cloud and Mobile Computing Era	2000–present	Servers store data and software in remote cloud computing centers where they can be accessed by computers, smartphones, and other client devices linked to the Internet. Computer processing, data storage, and application development take place in cloud computing centers as services, so that subscribing companies can maintain their IT infrastructures remotely.

IT Infrastructure and Its Evolution

(5 of 6)

- **Client/server computing**

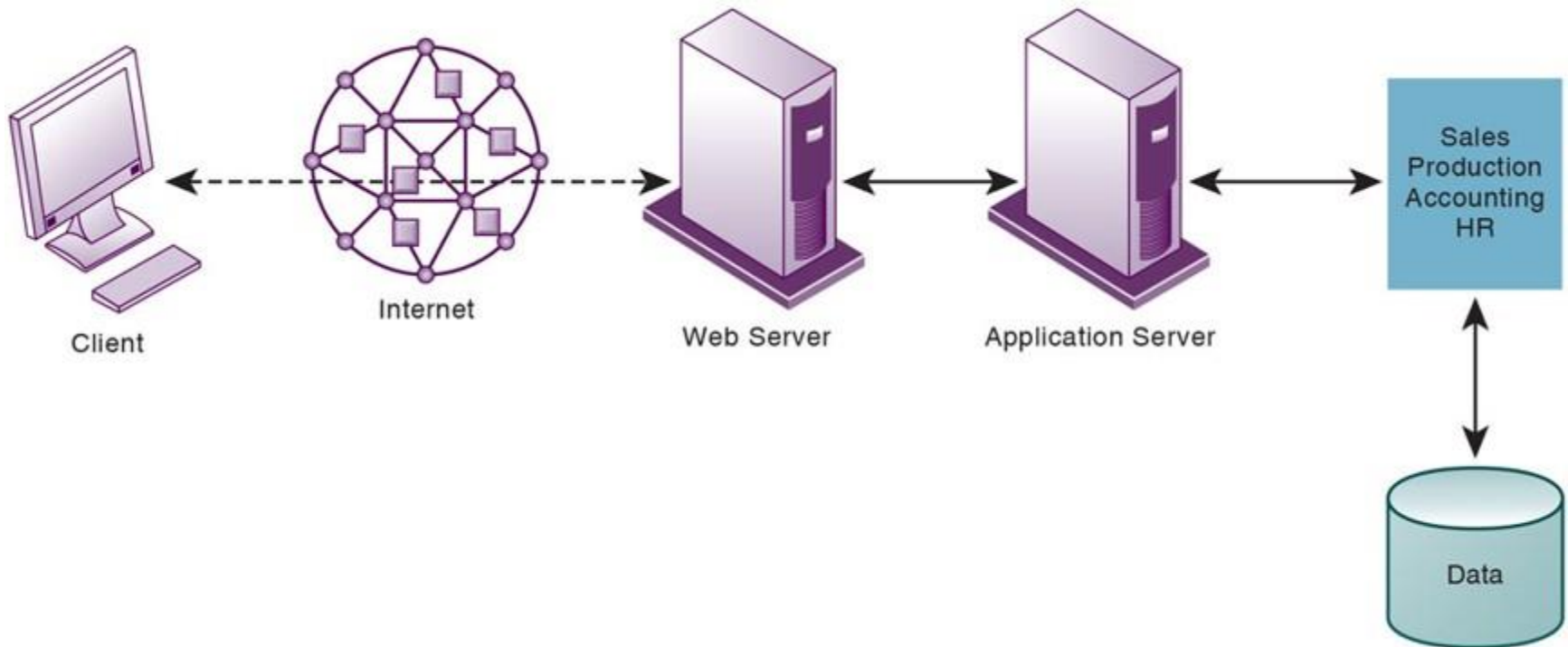
- The computing architecture most widely used today
- A model of computing in which a client is connected in a network with one or more servers
 - **Client:** Various types of devices that can request access to services or resources
 - **Server:** Computers that are dedicated to performing common functions that the clients on a network need
 - File storage
 - Software applications
 - Printing
 - Internet access

IT Infrastructure and Its Evolution

(6 of 6)

- Web server
 - Responsible for locating and managing stored web pages
 - Delivers web pages to a client in response to a request for service
- Application server
 - Handles all application operations between a browser-based client and an organization's back-end business applications or databases

Figure 5.3 Multitiered (N-Tier) Client/Server Network



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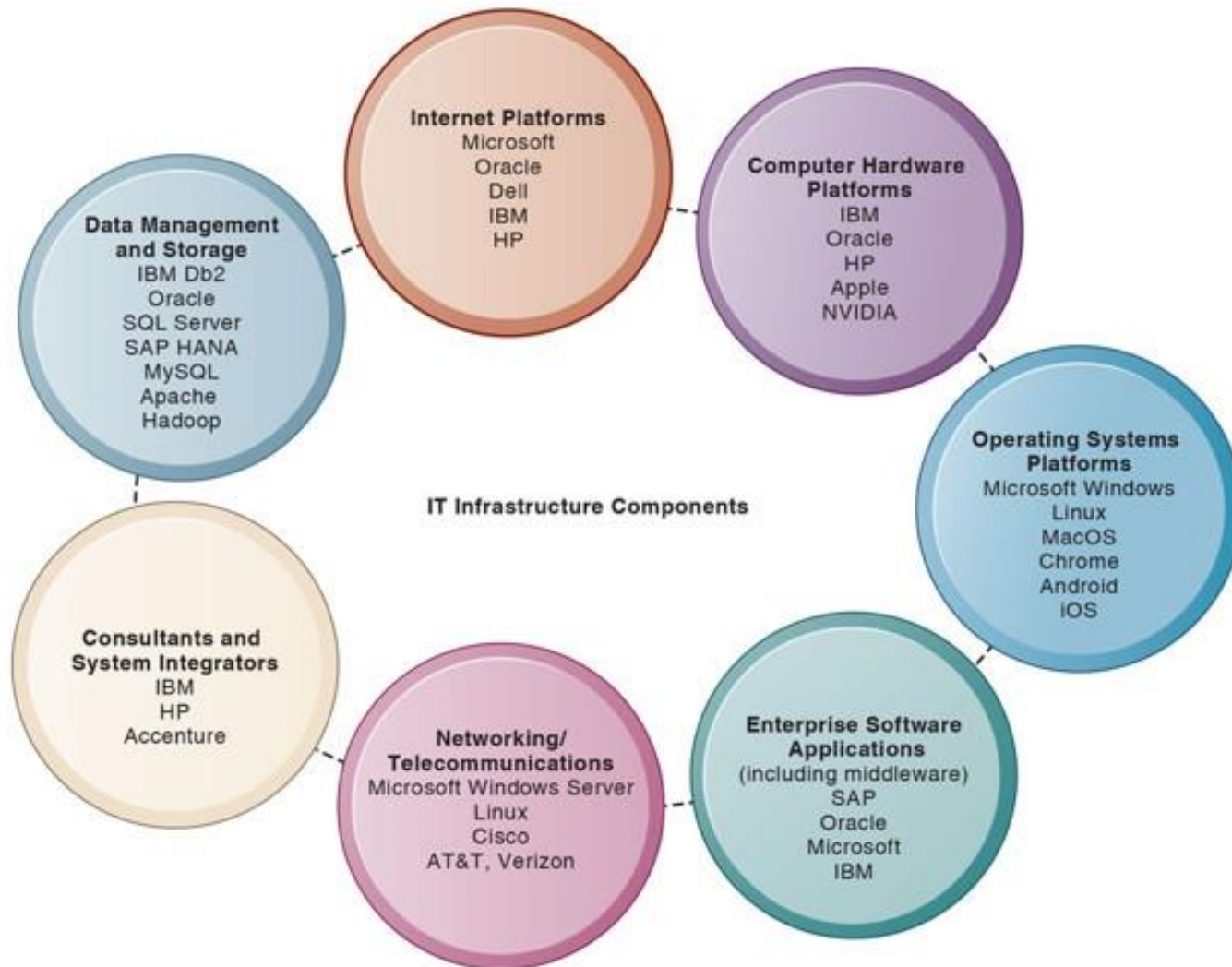
Components of IT Infrastructure

(1 of 5)

The components of IT infrastructure

1. Computer hardware platforms
2. Operating system platforms
3. Enterprise software applications
4. Data management and storage
5. Networking/telecommunications platforms
6. Internet platforms
7. Consulting and system integration services

Figure 5.4 IT Infrastructure



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Components of IT Infrastructure

(2 of 5)

- **Computer hardware platforms**

- Computer hardware provides technology for computer processing, and includes
 - Mainframes, servers, desktop computers, tablets, smart phones, as well as sensors and wearable devices such as smartwatches

Components of IT Infrastructure

(3 of 5)

- **Operating system platforms**

- Operating system

- System software that manages and controls activities of a computer

- Microsoft Windows operating system

- Over 68% of desktop computers and tablets use some form of this operating system

- Chrome OS

- Android

- iOS

- Multitouch

- Microsoft Windows Server

- UNIX

Components of IT Infrastructure

(4 of 5)

- **Enterprise software applications**

- An important component of IT infrastructure
- Largest providers: SAP and Oracle
- Middleware providers: IBM and Oracle

- **Data management and storage**

- Enterprise database management software: Responsible for organizing and managing a firm's data
- Can be efficiently accessed and used for operations and decision making
- The leading database software providers: IBM(Db2), Oracle, Microsoft (SQL Server), and SAP (HANA)

Components of IT Infrastructure

(5 of 5)

- **Network operating software (NOS)**
 - Windows Server, Linux
- **Internet platforms**
 - Hardware, software, and services to support company websites
 - Web-hosting services
- **Consulting and system integration services**
 - Software integration means ensuring new infrastructure works with older, legacy systems (older system that continues to be used to avoid higher cost systems)
 - Ensuring new elements of the infrastructure work together

The Mobile Platform, Apps, and BYOD

- **The Mobile platform**

- Smartphones; small, lightweight subnotebooks; tablets; car infotainment systems; wearable devices; and digital e-book readers

- **Apps**

- A software program that can be accessed via the Internet

- **Consumerization of IT and BYOD**

- Forces businesses and IT departments to rethink how IT equipment and services are acquired and managed
- Bring your own device (BYOD) into the workplace

Quantum Computing and Green Computing

- Two contemporary infrastructure trends
 - **Quantum computing**
 - Uses quantum physics to represent and operate on data
 - Dramatic increases in computing speed
 - **Green computing (or green IT)**
 - Refers to practices and technologies for designing, manufacturing, using, and disposing of computers and other IT equipment in order to minimize environmental impact

Quantum Computing - Benefits

Quantum Computing Benefits	Description
Speed	Quantum computers are up to 158 times faster than conventional computers.
Ability to solve more complex problems	Quantum computers are designed to perform many complex calculations simultaneously. Useful for developing decryption technologies.
Simulations	Quantum computers can run more complex simulations and can simulate more intricate systems than classical computers.
Optimizations	Quantum computing's ability to process huge amounts of complex data can make artificial intelligence and machine learning much more powerful.

Quantum Computing - Limitations

Quantum Computing Limitations	Description
Very heavy energy and resource requirements	Quantum computers require a significant amount of energy and cooling to run properly. Quantum computing cooling systems must keep a superconducting processor at a super-cooled temperature to allow electrons to flow freely. (IBM cools its quantum system to -459 degrees Fahrenheit.)
Error correction	Qubits can't use conventional error correction, which is critical in quantum computing. A single error in a calculation can affect the validity of an entire computation. Researchers are working on quantum error-correction algorithms.
Interference	The slightest disturbance in a quantum system can cause a quantum computation to collapse. A quantum computer must be totally isolated from all external interference during the computation phase.
Output observance	Retrieving output data after a quantum calculation is complete risks corrupting the data. Database search algorithms are being developed to address this problem.
Security	Quantum computing poses a profound threat to current classical encryption protocols that enable virtually all digital transactions. Over the next several years, widely used data encryption mechanisms, such as public key cryptography, could become vulnerable.

Examples of Green Computing Practices

Green practice	Description
Energy-efficient hardware	Energy-efficient computers and data centers that reduce power consumption and carbon emissions. This includes energy-efficient computing devices of all sizes, including smartphones, as well as energy-efficient microprocessors such as AMD Ryzen processors that minimize power consumption
Eliminating e-waste	Recycling and disposal of electronic waste according to federal, state, and local regulations
Optimizing heating, cooling, and electrical systems	Using IoT sensors and AI to collect and analyze information about power usage in a data center or a business and autonomously manage and optimize heating, cooling, and electrical systems
Using alternative energy sources and cooling methods	Using alternative energy sources, such as wind and hydroelectric power, as well as geothermal cooling and other new methods of cooling data centers.
Using and developing sustainable software	Includes sustainability and ESG reporting software, such as carbon accounting and carbon management software, and environmental management software. Also includes sustainable software engineering practices for defining, building, and running sustainable software applications that require less energy to run, using green coding practices such as optimizing algorithms, reducing code complexity, and minimizing the use of computing resources
Supporting remote work	Remote work decreases the number of people commuting to and from work, and the carbon emissions from their travel. It also reduces emissions from maintaining buildings for on-site work.
Server virtualization	Enabling multiple virtual machines to operate on a single physical server, reducing the need for multiple servers. More efficient use of server capacity decreases energy usage and CO ₂ emissions.

Virtualization and Cloud Computing

(1 of 6)

- **Virtualization**

- Computing resources not restricted by physical configuration or geographic location
- Enables a single physical resource (such as a server or a storage device) to appear to the user as multiple logical resources
- Reduces hardware and power expenditures
- Facilitates hardware centralization

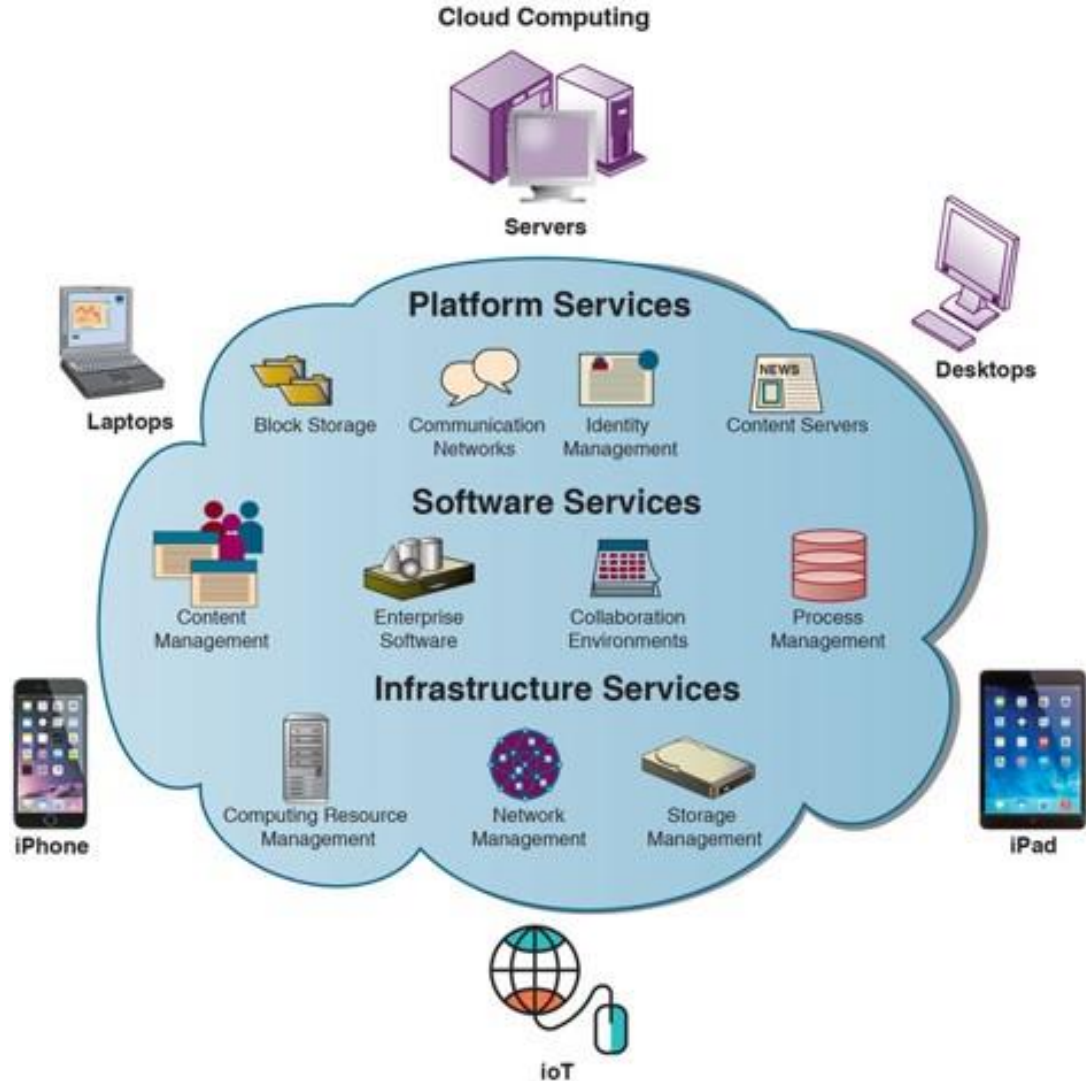
Virtualization and Cloud Computing

(2 of 6)

- **Cloud computing**

- A model of computing in which computer processing, storage, software, and other services are provided as
 - A shared pool of virtualized resources over a network, primarily the Internet

Figure 5.5 Cloud Computing Platform



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Virtualization and Cloud Computing

(3 of 6)

- Cloud computing can be used to provide three different types of services
 - **Infrastructure as a service (IaaS)**
 - e.g., Amazon uses the spare capacity of its IT infrastructure to provide a broadly based cloud environment - including Simple Storage Service (S3) for storing customers' data and Elastic Compute Cloud (EC2) service for running their applications.
 - **Software as a service (SaaS)**
 - e.g., Google Workspace, Microsoft 365, and related software services for desktop productivity and collaboration.
 - **Platform as a service (PaaS)**
 - e.g., Microsoft's PaaS tools and services for software development and testing as part of its Azure cloud services; or Salesforce's Salesforce Platform.

Figure 5.6 Amazon Web Services



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Virtualization and Cloud Computing

(4 of 6)

- There are several different cloud computing models
 - **Public cloud**
 - On-demand computing
 - **Private cloud**
 - managed by the organization or a third party and may be hosted either internally or externally
 - **Industry-cloud platform**
 - E.g., Salesforce's array of industry-specific clouds, including industry clouds for government, education, health, energy, and financial services.
 - **Hybrid cloud**
 - Companies use own infrastructure for most essential core activities and adopt public cloud computing for less critical systems or for additional processing capacity during peak business periods

Table 5.4**Cloud Computing Models Compared**

Type of Cloud	Description	Managed by	Uses
Public cloud	Third-party service offering computing, storage, and software services to multiple customers and that is available to the public	Third-party service providers	Companies without major privacy concerns Companies seeking pay-as-you-go IT services Companies lacking IT resources and expertise
Private cloud	Cloud infrastructure operated solely for a single organization and hosted either internally or externally	In-house IT or private third-party host	Companies with stringent privacy and security requirements Companies that must have control over data sovereignty
Hybrid cloud	Combination of private and public cloud services that remain separate entities	In-house IT, private host, third-party providers	Companies requiring some in-house control of IT that are also willing to assign part of their IT infrastructures to a public cloud

Virtualization and Cloud Computing

(5 of 6)

- All of the major cloud computing vendors offer a wide range of AI services
 - Amazon (AWS)
 - Microsoft Azure
 - Google Cloud Platform (GCP)
 - Salesforce

Table 5.5**Major Cloud AI Services**

Cloud Computing Vendor	Capabilities
Amazon Web Services (AWS)	Amazon SageMaker offers fully managed machine learning services. Includes tools for building, training, and deploying machine learning models, as well as pre-built functionalities for computer vision, natural language processing, and more.
Microsoft Azure	Microsoft Azure's AI platform focuses on enterprise solutions. Offerings include pre-trained models, cognitive services, and developer tools. Excels in integration with existing Microsoft products.
Google Cloud Platform (GCP)	GCP leverages Google's expertise in machine learning research. It is known for strong AI tools like AutoML, which simplifies model building, and TensorFlow, a popular open-source machine learning library.
Salesforce	Salesforce specializes in AI services specifically designed for the Salesforce CRM platform. These services focus on sales and customer service automation, such as lead scoring and sentiment analysis

Virtualization and Cloud Computing

(6 of 6)

- **Edge computing**

- A method of optimizing cloud computing systems by performing some data processing on a set of linked servers at the edge of the network, near the source of the data
 - Useful when sensors or other IoT devices do not need to be constantly connected to a central cloud
 - Reduces delays in transmitting and processing large quantities of data needed for AI applications

Contemporary Software Platforms

- Four major themes in contemporary software platform evolution
 - Open-source software
 - Java, HTML, and HTML5
 - Web services
 - Software outsourcing and cloud software services

Open-Source Software

- Software produced by a community of thousands of programmers around the world
 - Free
 - Can be modified by users
 - All works derived from the original code must be free
 - Not restricted to any specific operating system or hardware technology
 - e.g., Linux operating system, Apache HTTP web server, Mozilla Firefox web browser, and Apache OpenOffice desktop productivity suite.
 - Also, Google's Android mobile operating system and Chrome web browser are based on open-source tools.

Software for the Web: Java, HTML, HTML5

(1 of 2)

- Java

- An operating system-independent, processor-independent, object-oriented programming language
- Created by Sun Microsystems

Software for the Web: Java, HTML, HTML5

(2 of 2)

- **Web browser**

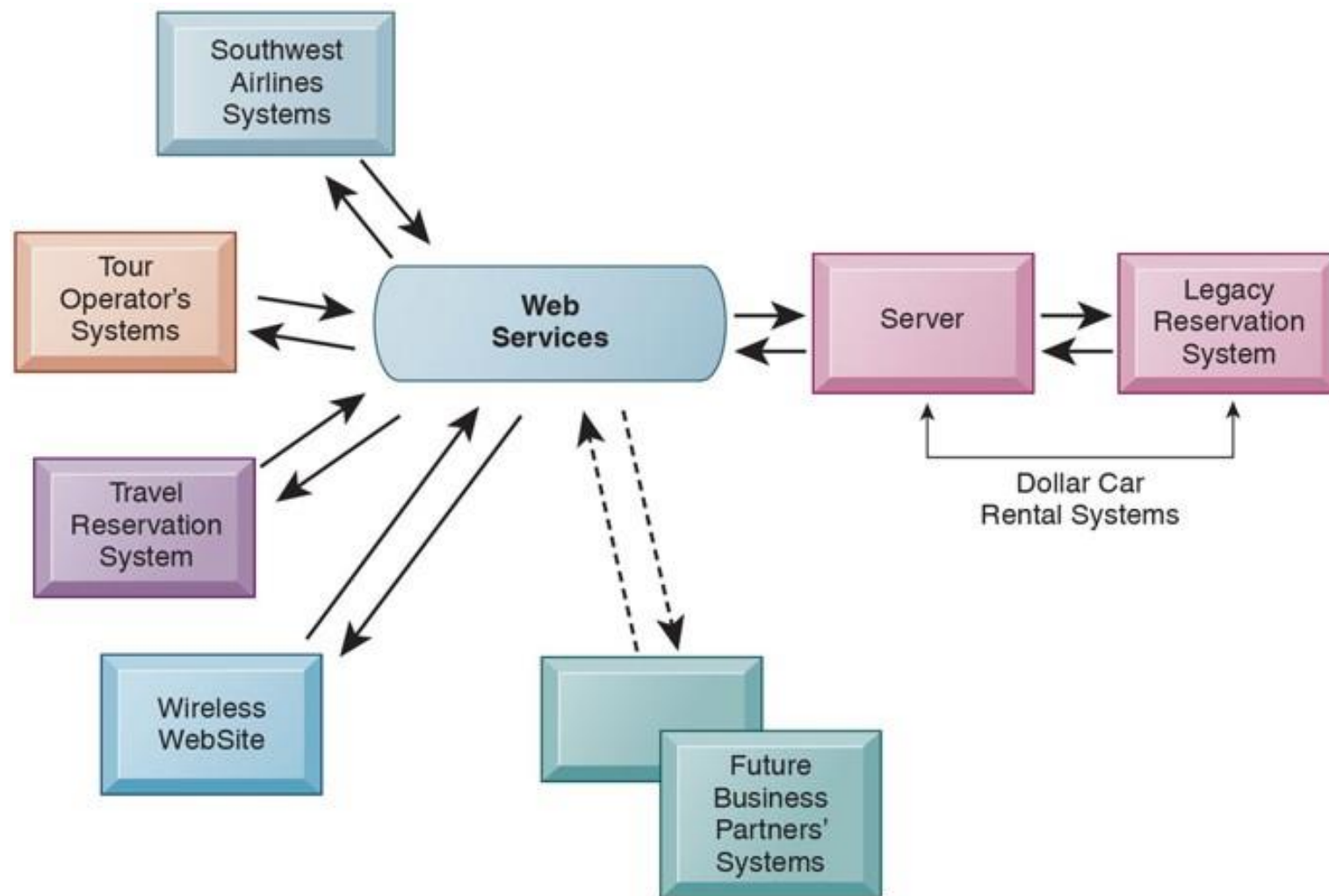
- Easy-to-use software tool with a graphical user interface
- Displays web pages
- For accessing the web and other Internet resources
- **Hypertext Markup Language (HTML)**
 - Page description language for specifying how text, graphics, video, and audio are placed on a web page
- **HTML5**
 - It is possible to embed images, audio, video, and other elements directly into a document without processor-intensive add-ons.
- **JavaScript**
 - Another core technology for making web pages more dynamic

Web Services

- Refer to a set of loosely coupled software components that exchange information with each other
 - Using universal web communication standards and languages
 - **Extensible Markup Language (XML)**
 - Foundation technology for web services
 - Includes "data tagging"

Plain English	XML
Subcompact	<AUTOMOBILETYPE= "Subcompact">
4 passenger	<PASSENGERUNIT= "PASS">4</PASSENGER>
\$16,800	<PRICE CURRENCY= "USD">\$16,800</PRICE>

Figure 5.7 How Dollar Car Rental Uses Web Services



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Alternative Ways of Obtaining Software

(1 of 4)

- Three external sources for software
 - Software packages from a commercial software vendor
 - Cloud-based software services and tools (SaaS/PaaS)
 - Outsourcing custom application development to an external vendor

Alternative Ways of Obtaining Software

(2 of 4)

- **Commercial software**

- Prewritten commercially available set of software programs that eliminates the need for a firm to write its own software programs
- Typically, software is downloaded from a vendor's website or may use the software as a cloud service delivered over the Internet
 - Pay a subscription fee

Alternative Ways of Obtaining Software

(3 of 4)

- **Cloud-based software**

- Hosted on servers in data centers
- Service-level agreement (SLA)
 - Formal contract between customers and service providers
 - Defines specific responsibilities of the service provider and the level of service expected by the customer

Alternative Ways of Obtaining Software

(4 of 4)

- **Software outsourcing**

- Enables a firm to contract custom software development or maintenance to outside firms
 - Often operate offshore in low-wage areas of the world where there are many skilled IT specialists

IT Infrastructure for AI

- Developing and running AI applications require specialized hardware and software
 - Based on the size of the dataset
 - Based on the intended use of the application

Table 5.7

AI Hardware Components

Component	Description
Central Processing Unit (CPU)	AI workloads require a powerful CPU to handle general computing tasks.
Graphics Processing Unit (GPU)	GPUs excel at parallel processing, making them ideal for training large AI models.
Tensor Processing Unit (TPU)	TPUs, developed by Google, are specialized chips optimized for machine learning tasks. They offer excellent performance for specific workloads.
System Memory (RAM)	Large datasets such as those in AI require very large amounts of system memory.
Data Storage and Management	Efficient data storage (databases, data lakes) and processing are essential for managing the large datasets used in AI training. Cloud storage solutions such as Amazon S3, Google Cloud Storage, and Azure Blob Storage are commonly used.

Table 5.8

AI Software Components

Component	Description
Operating system	Linux operating systems are often used because of their flexibility and compatibility with AI libraries. The open-source nature of Linux environments works well in complex installation and configuration processes required by many machine learning applications.
AI Frameworks and Libraries	Frameworks like TensorFlow, PyTorch, or Keras (which are all open source) provide tools for building and training AI models. Many developers are choosing open-source AI frameworks over proprietary software.
Version control	The GitHub platform allows software developers to create, store, manage and share their source code, with capabilities for version control.
Cloud services	Cloud platforms such as AWS, Azure, or Google Cloud are very scalable and often easier to deploy for AI than custom in-house systems.

AI and Edge Computing (Edge AI)

- Edge AI refers to the deployment of AI algorithms and AI models
 - Directly on local edge devices such as smartphones, sensors, or IoT devices

IT Infrastructure Management Challenges

(1 of 3)

- Creating and managing a coherent IT infrastructure raises multiple challenges
 - Dealing with platform and technology changes (including cloud and mobile computing)
 - Management and governance
 - Making wise infrastructure investments

IT Infrastructure Management Challenges

(2 of 3)

- As firms grow, they often quickly outgrow their infrastructure
 - Scalability needed
 - Ability of a computer, product, or system to expand to serve a large number of users without breaking down
- Mobile device management (MDM) software
 - Monitors, manages, and secures mobile devices that are deployed across multiple mobile service providers
 - Across multiple mobile operating systems being used in the organization

IT Infrastructure Management Challenges

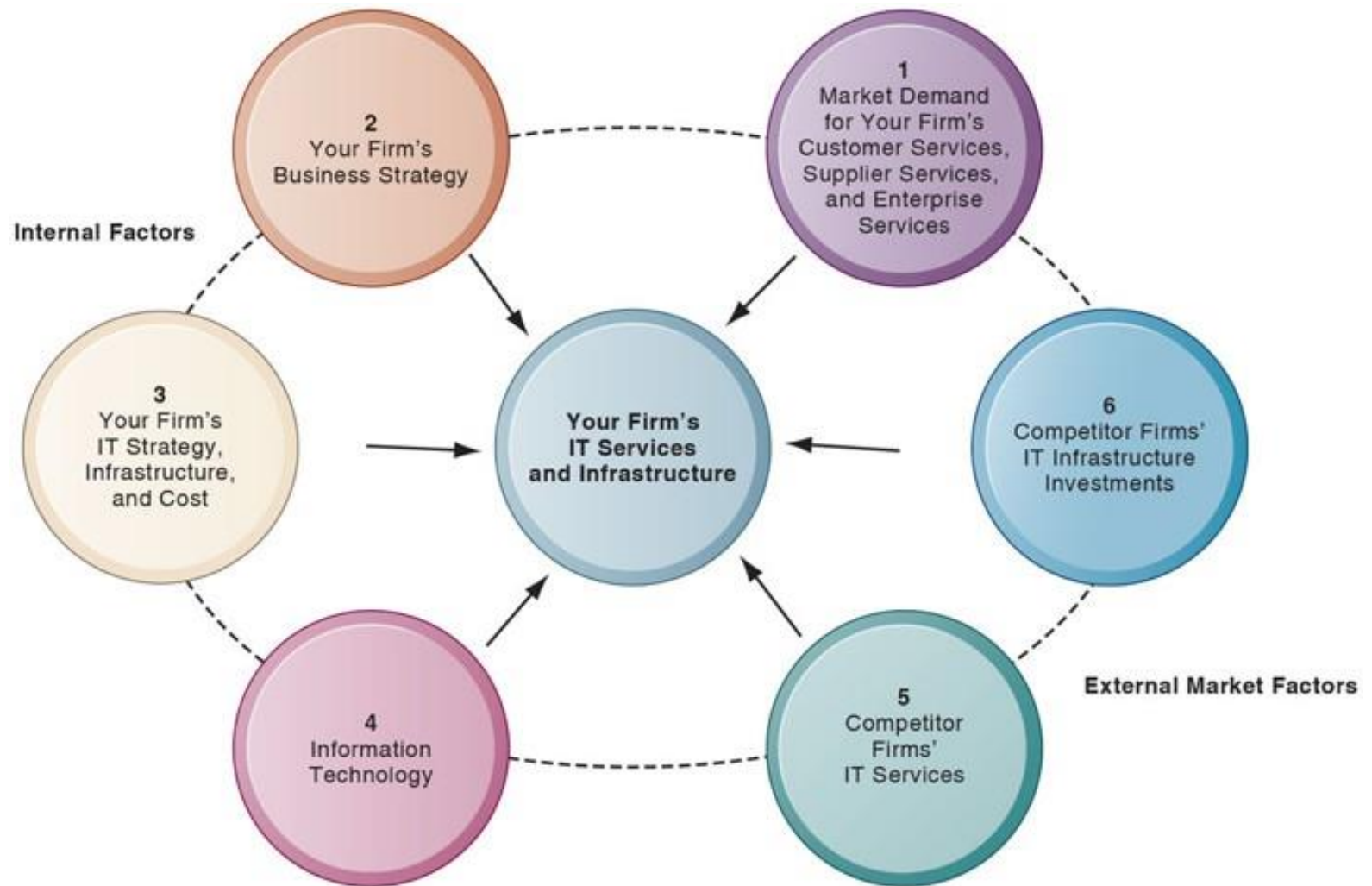
(3 of 3)

- IT infrastructure is a major investment for the firm
 - How much should the firm spend on infrastructure?
 - The rent-versus-buy decision
 - Cloud computing is often a low-cost way to increase scalability and flexibility
 - Total cost of ownership (TCO)

Table 5.9**Total Cost of Ownership (TCO) Cost Components**

Infrastructure Component	Cost Components
Hardware acquisition	Purchase price of computer hardware equipment, including computers, storage, printers, and mobile devices
Software acquisition	Purchase or license of software for each user or for the organization as a whole
Installation	Cost to install computers and software
Training	Cost to provide training for information systems specialists and end users
Support	Cost to provide ongoing technical support, help desks, and so forth
Maintenance	Cost to upgrade the hardware and software
Infrastructure	Cost to acquire, maintain, and support related infrastructure, such as networks and specialized equipment (including storage backup units)
Downtime	Cost of lost productivity if hardware or software failures cause the system to be unavailable for processing and user tasks
Space and energy	Real estate and utility costs for housing and providing power for the technology

Figure 5.8 Competitive Forces for IT Infrastructure



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